AUDIT-SID: Artifact-Level Diagnostics for Semantic-ID Tokenizers in Generative Recommendation

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Abstract

Semantic-ID tokenizers have become an interface layer in generative recommendation: they turn items into discrete addresses that a sequence model can generate. Aggregate Recall@K or NDCG, however, often gives little information about the address space itself. Codebook collapse, full-code collisions, behaviorally weak prefixes, tail compression, and large prefix fan-out can remain hidden until after expensive generator training. We present AUDIT-SID, a mapping-first diagnostic toolkit for item-to-SID tokenizer artifacts. AUDIT-SID normalizes item mappings, metadata, interactions, and optional generator outputs, then reports utilization, collision profile, semantic-collaborative alignment, head-tail allocation, and structural cost proxies. On a same-item Musical case study with 23,742 items, a GRID/RQ-KMeans-style feature-text export has 3,749 unique full SIDs and a 0.9769 full-collision rate, while a bounded ReSID/GAOQ export is collision-free under its exported mapping; the GRID collision rate remains 0.9751–0.9769 across three seeds. Controlled stressors further show that raw collision volume, interaction-qualified collision risk, tail capacity, and active-prefix cost are distinct artifact signals. The contribution is a reusable artifact audit suite, not a new tokenizer or a generative-recommender leaderboard.

CCS Concepts

• **Information systems** → **Recommender systems**; *Retrieval models and ranking*.

Keywords

semantic IDs, generative recommendation, tokenizer diagnostics, recommender systems

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1 Introduction

Semantic-ID (SID) tokenization has become a practical design point for generative recommendation. Systems such as TIGER map items into discrete code sequences so that a sequence model can generate candidate identifiers rather than score every catalog item directly [21]. This shift has made the tokenizer itself a first-order artifact. Recent work has expanded the space quickly: residual quantization frameworks, recommender-native encoding, collaborative tokenization, differentiable tokenization, non-uniform capacity, qualified collisions, variable-length interfaces, and drift-aware refreshes all change the item address space exposed to a generator [3, 6–8, 11, 15, 24, 27].

The evaluation practice has not caught up with this artifact boundary. Prior recommender tooling has argued for behavioral tests beyond NDCG-style aggregates [4], but SID tokenizers introduce a distinct artifact: the generated item address space. A tokenizer can support a usable downstream Recall@K while still hiding failure modes that are hard to diagnose from aggregate ranking metrics: underused codebooks, repeated full codes, semantically neat but behaviorally poor prefixes, tail items sharing too little capacity, or prefix structures that increase serving cost. These issues matter because SIDs are not merely features. They define the address space exposed to the generator.

This gap is especially awkward for resource reuse. A new tokenizer repository may expose checkpoints, item mappings, codebooks, or only intermediate feature files. Downstream users then need to answer basic engineering questions before training a generator: do all catalog items have valid codes, are many items aliased to the same full code, are collisions concentrated in the tail, and do prefix neighborhoods reflect user co-occurrence rather than only metadata taxonomy? Today these checks are usually reimplemented ad hoc inside each paper. The resulting evidence is difficult to compare because each method reports a different mixture of codebook usage, downstream ranking, and qualitative examples.

We present AUDIT-SID, a resource for auditing item-to-SID tokenizer artifacts before or alongside downstream recommendation evaluation. The goal is deliberately narrower than proposing a new tokenizer. AUDIT-SID asks a concrete question: given a public tokenizer artifact that maps items to code sequences, what can be said about its capacity use, collisions, behavioral alignment, head-tail allocation, and deployment cost without retraining a full generator?

The resource contribution is threefold. First, AUDIT-SID defines a small mapping-first interface for SID assignments, item metadata, interaction histories, and optional generator outputs. Second, it implements D1–D5a diagnostics for utilization, collisions, semantic-collaborative alignment, head-tail capacity, and structural serving-cost proxies, with D6 churn support for continual-tokenizer settings. Third, it demonstrates the audit on public GRID/RQ-KMeans-style, ReSID/GAOQ, sanity, and controlled-stressor artifacts. The same-item Musical case study and stressor table support diagnostic claims: they separate full-code collision pressure, interaction-qualified collision risk, collaborative-prefix recovery, tail capacity, and structural prefix fan-out before a downstream generator is trained. The paper follows the CIKM Resource four-page constraint, where appendices count toward the limit and detailed logs are therefore placed in the online artifact [5].

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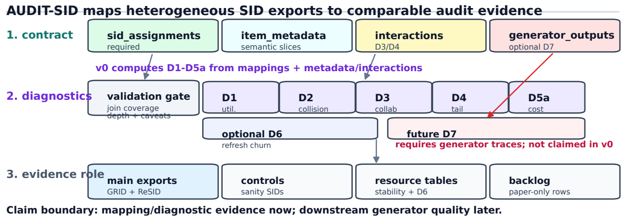

Figure 1: AUDIT-SID turns heterogeneous SID tokenizer exports into comparable audit artifacts. Exports are normalized into a mapping-first contract, validated for joinability and provenance, and summarized through D1–D5a diagnostics. D6 is optional churn evidence; D7 is enabled only when generator outputs or beam traces are provided.

2 Resource Scope and Interface

AUDIT-SID treats a tokenizer export as an inspectable resource artifact, not as an opaque component of a trained recommender. The minimum input is a table `sid_assignments(item_id, sid_0, ..., sid_L)`. Optional tables provide metadata, interactions, refresh pairs, and generator outputs. This mapping-first contract lets adapters normalize heterogeneous repositories while keeping the audit independent of a particular training loop. Figure 1 summarizes the resource workflow.

Artifact contract. The required contract has three checks: each row has a stable item key, every SID level is a discrete token, and the diagnostic item universe is joinable to metadata and interaction slices. Metadata enables semantic/category slices; interactions enable collaborative-neighborhood and head-tail diagnostics. The `generator_outputs` table is optional because many public tokenizer releases expose item codes without trained generator traces.

Evidence roles. AUDIT-SID separates rows by what they can support. A named-method row comes from a public method path and can support method-facing artifact claims. A controlled stressor is deliberately synthetic or method-inspired and can test whether a diagnostic reacts to a mechanism, but it is not method coverage. Optional rows exercise extensions such as drift or non-Amazon schemas. Literature-only rows motivate the design space but are not loaded as runnable evidence.

Validation before metrics. Adapters are refused as evidence until they pass item-count agreement, missing-code, duplicate-key, depth-consistency, and provenance checks. This guard is necessary because public SID repositories may expose checkpoints, item mappings, codebooks, intermediate feature files, or only partial releases. The manifest records each row's evidence role and caveats before any D1–D7 number is emitted.

Prior recommender resources such as RecList and Elliot motivate diagnostic and reproducible evaluation beyond a single aggregate metric [1, 4]. AUDIT-SID applies that resource logic to a newer artifact boundary: the item address space consumed by generative recommenders.

3 Diagnostic Design and Method Space

SID papers now optimize different artifact properties: learned item indexing and TIGER-style generative retrieval established the interface [9, 21], while later work stresses collaborative alignment, end-to-end or differentiable tokenization, representation-source sensitivity, code-assignment diversity, collision qualification, variable-length interfaces, drift, and deployment surfaces [2, 3, 7, 8, 12, 17, 18, 20, 23, 26, 27]. Table 1 organizes this space by diagnostic facet rather than by leaderboard rank.

D1–D4. D1 reports per-level usage, prefix counts, imbalance summaries, and dead or underused code indicators. D2 measures duplicate full SIDs and prefix collisions; it is a collision profile, not a causal harm estimate. A bounded interaction-qualified controller is included because collision-aware work argues that collisions are not uniformly harmful [8]. D3 asks whether prefix neighborhoods recover item co-occurrence neighbors, keeping metadata purity as auxiliary context rather than as collaborative quality. D4 splits items by popularity and measures whether full-code capacity and prefix structure are allocated differently across head, mid, and tail items.

D5a–D7. D5a reports structural cost proxies: SID length, prefix fan-out, duplicate codes, and trie-like expansion pressure. These are not serving latencies; long or variable SID methods make this distinction important [3, 25]. D6 computes churn between tokenizer refreshes for drift-aware settings [2, 6]. D7 is an interface hook for generator behavior, such as invalid paths, next-token entropy, and duplicate generated candidates; it requires `generator_outputs` and is not claimed by the current evidence.

4 Demonstration and Findings

The demonstration evaluates the resource interface rather than ranking tokenizer systems. It combines two public SID export paths with controlled rows that calibrate the diagnostics. GRID supplies the open RQ-KMeans-style path; ReSID supplies the GAOQ line and processed Musical artifacts. Sanity tokenizers and stressors are included only when they clarify what a diagnostic can and cannot identify.

Case-study protocol. The paper-facing case study uses the same 23,742 Musical items for both rows. The GRID row is a controlled feature-text export: ReSID processed feature text is embedded and passed through the official GRID-style residual k-means module. The ReSID row is a bounded FAMAE-to-GAOQ export on the same item universe. This modest protocol gives reviewers a same-item diagnostic contrast while avoiding the stronger, unsupported claim that the paper reproduces every training choice of TIGER, GRID, or the full ReSID pipeline.

Table 2 compares two exports on the same item set. The GRID row is intentionally labeled as feature-text: it uses processed feature text from the ReSID artifact path and the official residual quantization module, not a faithful raw-text TIGER reproduction. Under that controlled setup, the row exposes high collision pressure and limited tail capacity. The ReSID/GAOQ row exports one full code per item in this bounded run, yielding zero full-code collisions and higher D3-L1 collaborative prefix recovery. Across seeds 42–44, the GRID duplicate-SID rate remains 0.8327–0.8421 and full-collision

Table 1: Facet coverage matrix. 'M' denotes main PDF evidence, 'A' artifact repository evidence, 'H' interface hook, and '–' no v0 evidence. The table is coverage context, not a claim that all anchor methods are reproduced.

Facet	Anchors	v0 role	D1	D2	D3	D4	D5a	D6	D7
A. Canonical residual SID	TIGER, GRID, RQ-KMeans [11, 21]	GRID feature-text	M	M	M	M	M	–	–
B1. Collaborative / predictability	ReSID, LC-Rec, CoST, LETTER, DiscRec [15, 16, 23, 26, 27]	ReSID bounded	M	M	M	M	M	–	–
B2. Collision / capacity	QuaSID, AdaSID, CARD, CapsID [3, 8, 18, 24]	stressors/backlog	A	A	–	A	A	–	–
B3. Ranking / retrieval	DIGER, ELIT, joint search-rec SID [7, 17, 19]	interface hook	–	–	H	–	H	–	H
B4. Bottleneck / interface	AsymRec, CapsID, Long SID/ACERec [3, 10, 25]	future target	–	–	–	H	H	–	H
C. Drift / staleness	DACT, SID staleness [2, 6]	optional smoke	–	–	–	–	–	A	–
D. Deployment / search	Snapchat SID, SID-Coord, unified search-rec [12, 13, 19]	motivation	–	–	–	–	H	–	H
R0/control layer	RecList, Elliot, sanity SIDs [1, 4]	controls	A	A	A	A	A	–	–

Table 2: Same-item Musical artifact profiles on 23,742 items. Higher unique full codes, D3 L1 co-occurrence prefix recall, and D4 tail unique-SID ratio are better; lower D2 full-collision rate is better. D5a prefixes are level-wise prefix counts, not latency.

System	Unique	D2 full	D3 L1	D4 tail	D5a prefixes
GRID feature-text	3,749	0.9769	0.0552	0.3695	64/3440/3749
ReSID bounded	23,742	0.0000	0.1535	1.0000	32/1280/23742

Table 3: Controlled stressor signals. These rows are not named-tokenizer evidence; they test whether D2, D4, and D5a react to specific artifact mechanisms.

Stressor	Metric	Signal 1	Signal 2
Qualified collision	co-occurrence lift	3.86x	1.19x
Capacity budget	unique ratio	1.000 head	0.028 tail
Variable depth	prefix-cost view	active	max-depth

rate remains 0.9751–0.9769, so the collision pressure is not a single-seed artifact.

Diagnostic findings. Tables 2 and 3 support three artifact-level findings. First, collision pressure is stable in the controlled GRID row, while the same interface ingests a collision-free ReSID mapping. The qualified-collision stressor then shows why raw collision volume is insufficient: collisions that overlap user co-occurrence have a 3.86x lift, while synthetic hash collisions have 1.19x. Second, collision-free addresses and collaborative-prefix alignment are different objectives. ReSID has no full collisions and higher D3-L1 than the GRID row, while a category-prefix control in the repository recovers more level-1 co-occurrence neighbors without being a learned tokenizer. Third, D4 and D5a separate capacity allocation from prefix structure: in the capacity stressor, head uniqueness reaches 1.000 while tail uniqueness remains 0.028, and variable-depth stressors show that active prefix cost need not match a fixed maximum-depth view.

Repository tables extend these checks without changing the paper claims: sanity rows calibrate D1–D5a, GRID scale rows test larger item counts, DACT rows exercise optional D6 churn, and MovieLens rows check non-Amazon schema portability. Table 2 remains the only method-facing case study in the main paper.

Table 4: Reviewer-facing artifact contract. The table lists what a clean checkout exposes; expensive metric rebuilds remain optional and are documented through manifests.

Asset	Reviewer check	Claim supported
Setup	inspect setup files	availability
Validator	run clean-check verifier	interface contract
Tables	match CSVs to paper numbers	Tables 1–3
Manifest	inspect row roles and caveats	claim boundary
Extra panels	optional local review	portability, D6

5 Availability, Reproducibility, and Limits

The artifact repository contains the AUDIT-SID Python package, adapter scripts, metric runners, generated CSV/Markdown/LaTeX tables, provenance logs, BibTeX/source-audit notes, an MIT license, and a reviewer quickstart. The reviewer entry point is the tag `audit-sid-cikm-resource-v0.1`, with a clean-checkout verifier for the published tables and figure. Large local artifacts are kept out of git and described through manifests so reviewers can separate public reproducibility assets from local caches.

The resource is intended to be used in three modes. A paper reviewer can run the clean-check verifier and compare published CSVs with the PDF tables. A method author can add an adapter that emits the normalized `sid_assignments` table, then receive D1–D5a reports without changing their training loop. A downstream recommender researcher can use the same diagnostics as a pre-training gate: artifacts with missing items, inconsistent depth, severe collisions, or weak collaborative-prefix recovery can be flagged before generator training consumes GPU time. These modes are deliberately lighter than a full benchmark suite, but they match the resource-track goal of making a reusable software artifact available at review time.

The evidence supports a conservative resource-demo claim. The toolkit does not reproduce TIGER end-to-end; the GRID Musical row is a controlled feature-text export. The ReSID Sports balanced GAOQ path was stopped after constrained k-means became CPU-bound, so the main ReSID evidence is the bounded Musical export. CARD remains a fallback/stressor path, not faithful CARD reproduction [24]. D2 includes an interaction-qualified controller but not causal harm; D3 is not proven monotonic with Recall@K or NDCG;

D5a is structural rather than measured latency; and D6 is optional drift evidence.

AUDIT-SID audits tokenizer artifacts; it is not a new SID tokenizer, a complete generative-recommender evaluation, or an industrial online-impact study. Artifact diagnostics can reveal pressure points before generator training, but they do not replace ranking evaluation. Future versions should add causal collision-harm validation, generator-output diagnostics, stronger same-dataset method coverage, and unified search/recommendation checks [14, 19]. Industrial SID deployment and ranking studies further motivate treating these artifact properties as engineering objects [12, 13, 22].

New adapters emit the same normalized tables, declare their evidence role, and record feature or checkpoint provenance. Versioned manifests and fixed-name "latest" reports preserve stable entry points and timestamped provenance. Thus the Musical case study is not a ranking of GRID against ReSID; it shows how one artifact contract turns collisions, collaborative prefix recovery, tail allocation, and prefix fan-out into separately checkable review objects.

The extension path is also explicit. A new named tokenizer improves method coverage only when it exports item-level codes that pass the validator and can be joined to metadata or interactions. A controlled stressor improves metric calibration but does not count as method coverage. A generator trace activates D7, but without such traces AUDIT-SID reports only mapping-level and diagnostic claims. This separation is what keeps the artifact useful as the SID literature changes: new methods can be added without rewriting the diagnostics, while claims remain tied to the evidence role of each row.

GenAI Usage Disclosure

The authors used generative AI tools for drafting assistance, code-review support, and experiment-log organization. The reported claims, tables, and citations were checked against the artifact repository and cited sources during paper preparation.

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